IP Addressing, Subnetting, CIDR, and Fragmentation

Lecture 21

Computer Networks and Distributed Programming

Where the bits inside a prefix come from

- We left the forwarding table settled: a row is a prefix, and when several rows match, the longest one wins.
- What was never said is where the prefixes come from — who decides that four billion addresses are grouped in *that* way.
- Nothing in the hardware forces it. Addresses could be handed out one at a time, in the order machines are switched on.
- If they were, no router could hold a table. The whole arrangement rests on addresses being allocated in blocks.
- Today: how an address is built, how a network is cut into subnets, how blocks are handed out — and what happens to a datagram too large for the next link.

The Internet Protocol

- IP: the network-layer protocol that every host and every router runs. Two versions are in use, IPv4 and IPv6.
- IPv4 still carries almost all the traffic; IPv6 was designed to replace it, and we come to it when we run out of addresses.
- The service is the best-effort delivery we have already described: no connection, no acknowledgement, no guarantee.
- What IP adds is one thing — names for the endpoints. Every other function of this layer depends on that one.
- And it is hard for a mundane reason: the scheme must name billions of interfaces and still be searchable in nanoseconds.

Thirty-two bits written as four decimals

	Dotted-decimal	Binary
IP address	`193.32.216.9`	`11000001 00100000 11011000 00001001`

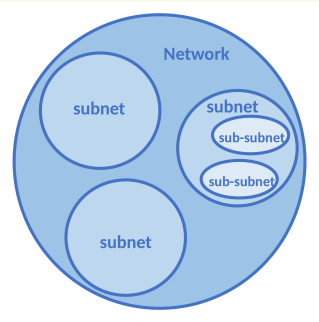
- An IPv4 address is 32 bits. Each of the four bytes is written in decimal and separated by a dot.
- The dotted form is a convenience for us. Nothing in the protocol uses it: every operation on addresses is an operation on bits.
- 2^{32}, about four billion addresses in all — a number that looked inexhaustible in 1981.
- Every interface in the global Internet must have an address that is globally unique, with one exception we will meet later.

An address names an interface, not a machine

- Hosts and routers attach to links. The boundary between a device and one of its links is an interface.
- A host normally has one. A router, by its job description, has at least two: it receives on one link and sends on another.
- IP requires every interface to have its own address. The address therefore belongs to the interface, not to the box behind it.
- A laptop on cable and wireless at the same time has two addresses; a router with three ports has three, and they need not resemble each other.
- It sounds like pedantry until we count the subnets of a real topology, where it is exactly what makes the count come out right.

Why addresses cannot be handed out at random

- Nothing prevents assigning addresses in the order the machines are bought. Two things go wrong at once.
- Nobody knows where anything is: an address carrying no trace of location tells a router nothing about the direction to send.
- The tables explode: without groups, every destination is its own row, and four billion rows per router is not an option.
- The way out is the one telephony already used: cut the space hierarchically, into networks and subnetworks.
- So an address is read in two parts — leftmost bits name the subnet, the rest name an interface inside it.



The mask says where the address splits

	Dotted-decimal	Binary
IP address	193.32.216.9	11000001 00100000 11011000 00001001
Subnet mask	255.255.255.0	11111111 11111111 11111111 00000000

- The number of bits in each part is not fixed, and the address does not carry it. It is stated separately, by a subnet mask.
- Ones over the subnet part, zeros over the interface part. Here 24 ones: three bytes of subnet, one byte of interface.
- The compact form puts the two together: 193.32.216.0/24, the number of leading bits that belong to the subnet.
- Same information, two notations. The slashed one is what routing protocols and administrators actually write.

A prefix is a range: minimum and maximum

Hold the subnet bits fixed and let the eight free bits run.

minimum	11000001 00100000 11011000 00000000	193.32.216.0
maximum	11000001 00100000 11011000 11111111	193.32.216.255

- The subnet is the whole interval between the two ends: $2^8 = 256$ addresses, and nothing outside it.
- Which is why the notation names a block by its lowest address and the prefix length: together they fix both ends.
- In general $32 - x$ free bits give 2^{32-x} addresses — 256 for a /24, 512 for a /23, 65 536 for a /16. Every block is a power of two.

Which masks are legal

255.255.255.0	11111111 11111111 11111111 00000000	mask (/24)
255.255.128.0	11111111 11111111 10000000 00000000	mask (/17)
255.224.0.0	11111111 11100000 00000000 00000000	mask (/11)
255.255.10.0	11111111 11111111 00001010 00000000	not a mask
63.255.255.0	00111111 11111111 11111111 00000000	not a mask

- The ones must be leftmost and contiguous: a mask with a hole in it, or one that does not start at the top, is not a mask.
- Contiguity is what makes the block an interval. Fix the first x bits, and the free bits below run from all-zeros to all-ones.
- With a hole, the matching addresses would be scattered through the space, and no comparison of leading bits would ever find them.

Same subnet or not

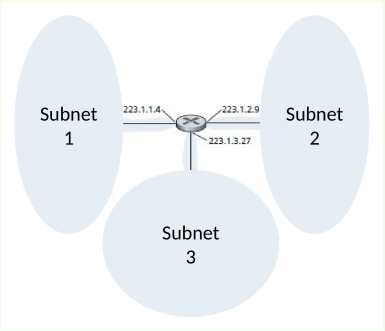
Example

IP 1	Mask	IP 2	
231.23.11.117	255.255.255.0	231.23.11.9	same subnet
10.54.32.1	255.255.128.0	10.54.60.203	same subnet
110.32.100.10	255.224.0.0	110.64.100.11	different
121.50.79.4	255.0.0.0	121.10.36.240	same subnet
121.50.79.4	255.0.0.0	120.50.79.5	different

- Two interfaces share a subnet when their addresses agree on the masked bits. The decimal digits are no guide at all.
- Third row: 32 and 64 already differ in the three bits that the /11 mask covers, although the first byte is the same.
- Hosts on different subnets still talk — through a router, which is also where an administrator can filter or block.

The router sits on the boundary

- What separates one subnet from another is a router: several ports, several interfaces, one address each.
- A router with one interface inside a subnet and one outside is that subnet's way out — its gateway.
- Because everything in and out passes through it, the box tends to collect other jobs: address translation, address assignment, filtering.
- Each of those is a protocol in its own right and we take them in turn. Here the router is simply what makes two subnets two.

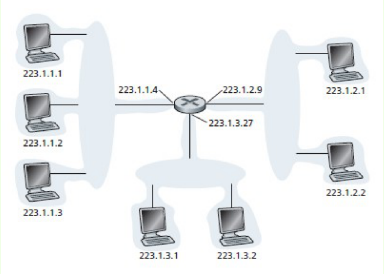


Detach the interfaces and count the islands

Subnet

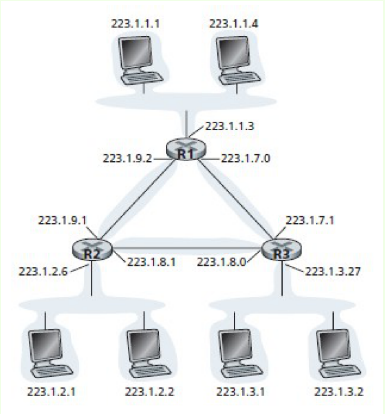
Detach every interface from its host or router; each isolated island that remains is a subnet.

- Three hosts *and* the router interface `223.1.1.4` form the left island.
- All four share 24 bits: the subnet is `223.1.1.0/24`, and a host added there must be a `223.1.1.xxx`.
- A subnet is not a cable and not a device: it is what is reachable without crossing a router.

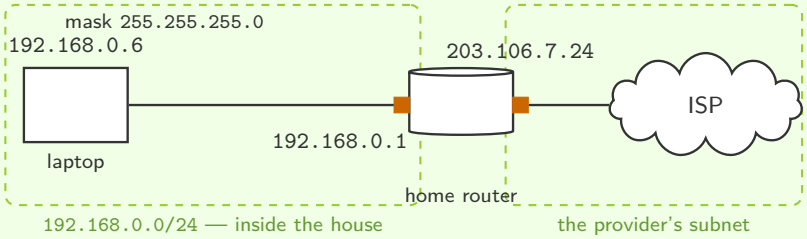


A router-to-router link is a subnet too

- Three routers, three interfaces each: one towards its hosts, two towards the other routers.
- The three islands of hosts are subnets as before: `223.1.1.0/24`, `223.1.2.0/24`, `223.1.3.0/24`.
- But detaching the interfaces leaves three more islands: every point-to-point link is a subnet of its own.
- `223.1.9.0/24`, `223.1.8.0/24` and `223.1.7.0/24` hold two interfaces and no host, and take a /24 each.
- Six subnets, not three: which is why blocks run short sooner than the host count suggests.



A house



- The home router has two interfaces, hence two addresses, one in each of the two subnets it joins.
- 192.168.0.0/16 is reserved for private use and is not globally unique: the same addresses are in millions of houses.
- How one public address serves all the devices behind it is the business of address translation, which we come to shortly.

Reading your own configuration

```
$ ip -4 addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    inet 217.149.127.10/26 brd 217.149.127.63 scope global eth0
       valid_lft forever preferred_lft forever
```

- `ip addr show` lists the interfaces with their configuration; `ipconfig` on Windows.
- The prefix is the interesting part: /26 is 26 ones, the mask 255.255.255.192, so only 6 bits are left.
- The subnet is `217.149.127.0/26`, running from `.0` to `.63`: sixty-four addresses, and no more.
- `mtu 1500` is the largest payload, in frames of up to 1518: we need it later.

Two addresses in every block are not yours

- The lowest address of a block, host bits all zero, names the subnet itself and is never given to an interface.
- The highest, host bits all one, is the broadcast address: a datagram sent there is delivered to every interface on the subnet.
- The configuration we have just read says so: `brd 217.149.127.63`. Of the 64 addresses of that /26, 62 can be assigned.
- So a /24 holds 254 usable addresses and not 256, and the smaller the block, the heavier those two weigh.
- 255.255.255.255: the limited broadcast address — every interface on *my own* subnet. Routers do not forward it.

Addressing the whole Internet

Cutting a campus into subnets is one thing. Assigning the space to billions of interfaces is another, and there are three constraints.

- Uniqueness: two interfaces with the same address and delivery has no meaning. Somebody has to arbitrate, globally.
- Table size: the tables stay small only if all the addresses of an organization can be summarized in one row.
- Exhaustion: the space is finite, and blocks handed out too generously are gone for good.
- Two schemes have been used: classful addressing, which is history, and classless addressing, which is what runs today.

Classful addressing

	Form	Example	Addresses
Class A	`a.b.c.d/8`	`10.x.x.x`	2^{24}, over 16 million
Class B	`a.b.c.d/16`	`10.10.x.x`	$2^{16} = 65\,536$
Class C	`a.b.c.d/24`	`10.10.10.x`	$2^{8} = 256$

- Until the nineties the boundary could only fall at 8, 16 or 24 bits: one, two or three whole bytes, and nothing in between.
- An organization with 300 hosts: a class C is 256 addresses, too few. So it was given a class B — and 65 236 addresses nobody else could ever use.
- The class B space drained fast, and the mismatch between what an organization needs and what it can be given was structural.

CIDR: the boundary moves

- CIDR: Classless Interdomain Routing, pronounced *cider*: the prefix may be of any length, `a.b.c.d/x`.
- The 300 hosts now take a `/23`: 512 addresses, 212 unused instead of sixty-five thousand.
- Prefix: the x leading bits, common to every address of the organization. Block: the set of addresses that share them.
- Nothing else changes: the mask is still ones then zeros, the block is still an interval. What went away is the constraint on x.
- And blocks are given out contiguously, which turns out to matter more than the addresses saved.

The same address lies in several blocks

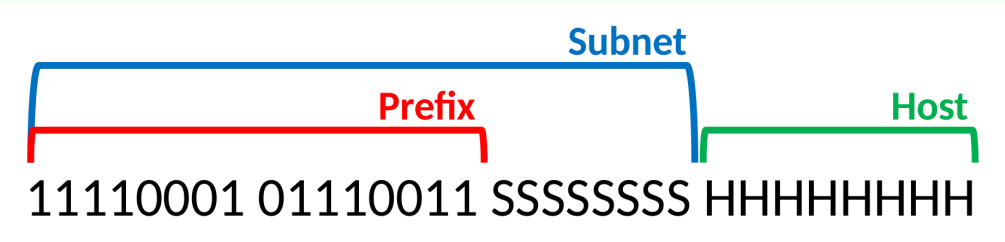
Which block does `10.10.10.15` belong to?

To `10.10.0.0/16`, to `10.10.10.0/24` and to `10.10.10.0/28`, all at once. An address on its own does not name a block: the prefix length is the missing half of the statement.

- The blocks are nested, one inside another, not laid side by side — and a smaller prefix length means a larger block.
- So a router may well find two rows that both match, and it already knows what to do: it keeps the longer prefix.
- Written with the length attached, `/16` and `/24` are two different claims about the same address, and nothing is ambiguous.

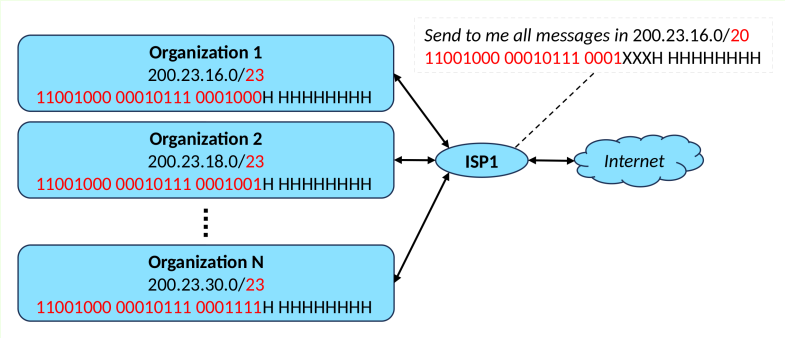
Subnets inside a block

- The bits below the prefix belong to the organization, which may cut them as it likes: routers outside never look at them.
- `241.115.0.0/16`: sixteen bits fixed by whoever assigned the block; the remaining sixteen are the organization's own business.
- It can spend eight of them on its internal subnets — `241.115.1.0/24`, `241.115.2.0/24`, and so on — and eight on hosts.



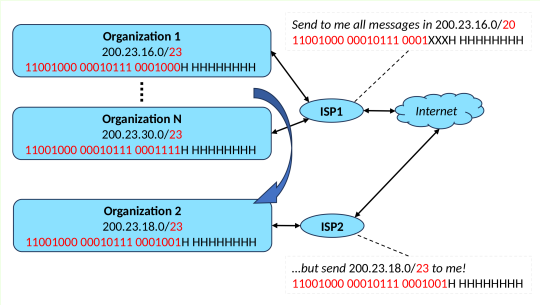
Inside the campus the routers use all 24 bits; outside, 16 are enough.

One announcement instead of N



- An ISP holding `200.23.16.0/20` cuts it into `/23` blocks, one for each organization it serves.
- They are consecutive: same first 20 bits, differing only in the three bits below.
- Address aggregation: one prefix announced for all of them, and one row in the world's tables instead of N.

When one organization moves



routing table	
200.23.16.0/20	L1
200.23.18.0/23	L2
⋮	⋮

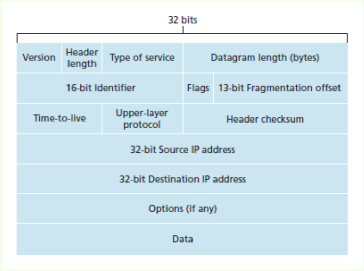
- Organization 2 changes provider and keeps its addresses: renumbering everything is the alternative, and nobody does it.
- Its block still sits inside ISP1's /20, so ISP2 announces the more specific /23. Both rows match, the longer wins: only that traffic takes L2 — aggregation dented, not broken.

Where a block comes from

- Ask your ISP, which carves your block out of the larger one it already holds. This is how the clustering happens in practice.
- Or go up: ICANN has ultimate responsibility for the address space and delegates to five regional registries — RIPE for Europe, then ARIN, APNIC, LACNIC, AFRINIC.
- ICANN also runs the DNS root servers and assigns domain names, so both namespaces the Internet uses answer to one authority.
- What is left is the last step: giving single addresses to the interfaces inside the block. A router's are configured by hand.
- A host's are handed out automatically by a protocol of its own, which we will meet with the rest of the plug-and-play machinery.

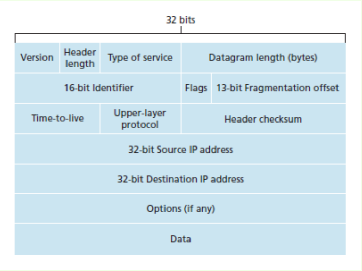
The IPv4 datagram: the first three words

- Version: 4 bits: which version, hence how to read all the rest.
- Header length: 4 bits: 20 bytes plus options, so the payload does not start at a fixed place.
- Type of service: 8 bits: classes of traffic; two of them carry the congestion notification we have seen.
- Datagram length: 16 bits: header plus data, up to 65 535 bytes, in practice below 1 500.
- Identifier, flags, offset: 32 bits, used only when a datagram has been cut in pieces.
- Time-to-live: 8 bits, decremented at each router, dropped at zero: what stops a routing loop.



The rest of the header

- Upper-layer protocol: 8 bits: who gets the payload — 6 is TCP, 17 is UDP. It is the glue between network and transport, as the port number is between transport and application.
- Header checksum: 16 bits over the header only, recomputed at every hop because the TTL changes.
- Source and destination: 32 bits each, written by the source and left alone on the way.
- Options: rare and variable in length — the reason the header length field exists. IPv6 dropped them.
- Data: a TCP or UDP segment, or a control message of this layer.



Two checksums over the same bytes

Why does a TCP segment travel under two checksums?

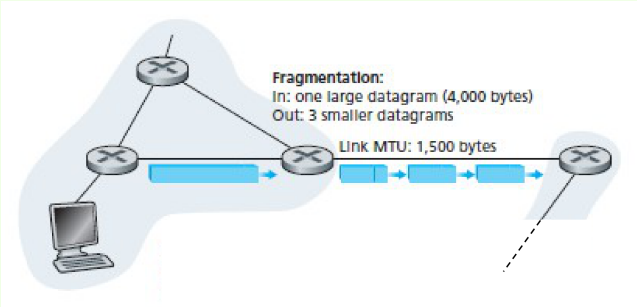
- They cover different things. IP checks its own header; TCP and UDP check the whole segment, payload included.
- A corrupted destination address is caught by the first. A bit flipped in the data is not, and needs the second.
- And the two layers are not welded together: IP carries things that are not TCP, and TCP could in principle run over something else.
- Neither layer may assume the other is doing the work — which is the general shape of the end-to-end argument.

The link decides the maximum size

- A datagram does not travel as such: it is handed down and carried inside a link-layer frame, and frames have a maximum size.
- MTU: maximum transmission unit, the largest payload a frame will carry. Ethernet 1 500 bytes, 802.11 2 272.
- The limits come from the hardware, from header fields, and from the wish that a single packet not hold the channel too long.
- IP allows datagrams up to 65 535 bytes, so a router may well receive on a wide link and have to send on a narrower one.
- Then: IP cuts the datagram into pieces that fit. Each piece is itself a datagram, and travels in a frame of its own.

How a fragment is labelled

- The router copies the identifier and both addresses into every piece: that is what marks them as parts of one datagram.
- Fragment offset: 13 bits: where this piece begins inside the original payload, counted in units of 8 bytes.
- More fragments: one bit, 1 in every fragment except the last, whose 0 says that the datagram ends here.



Four thousand bytes over a fifteen-hundred-byte link

Example

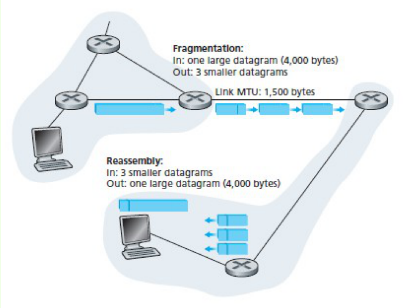
A 4 000-byte datagram, 20 of header and 3 980 of data, must go out on a link with MTU 1 500. Each fragment carries its own 20-byte header, so at most 1 480 bytes of data — and $1\,480 = 8 \times 185$.

	Data	Total	Offset	MF
fragment 1	1 480	1 500	0	1
fragment 2	1 480	1 500	185	1
fragment 3	1 020	1 040	370	0

- The offsets are byte positions divided by eight: 0, $1\,480/8 = 185$, $2\,960/8 = 370$.
- All three carry the same identifier and the same pair of addresses. Only the third says that the datagram ends there.

Reassembly happens at the destination

- Two designs were possible: put the pieces back together at the next router, or carry them to the destination host.
- Reassembling at each hop means buffering, waiting, deciding when to give up — and forcing all the fragments through one exit.
- IP chose the other one. Routers only cut; fragments are ordinary datagrams and are routed independently.
- The host recognizes them by same identifier, same source, same destination, and the offsets say where each one goes.
- It is done when it holds offset 0, a fragment with MF $= 0$, and no hole between them.



One lost fragment loses the datagram

Three fragments leave, one is dropped on the way. What arrives?

Nothing. The two that made it cannot be delivered upwards — the transport layer expects a whole segment — and are discarded when the reassembly timer expires.

- One per cent of fragments lost becomes three per cent of datagrams lost, and the sender retransmits the whole thing.
- Which is then fragmented again, over the same link, with the same odds. Add a header per piece and the work forced on the receiver.
- So fragmentation is a mechanism that works and that one would rather never use.

Don't fragment: path MTU discovery

- Modern senders avoid fragmentation rather than rely on it, and the header has a bit for it: DF, don't fragment.
- With DF set, a router that cannot forward the datagram drops it and returns an error to the source, naming the MTU it could have accepted.
- The source shrinks its packets and tries again; a narrower link further on triggers another error, and it shrinks again.
- Path MTU: the smallest MTU along the path. It costs round trips before any data arrives, and it changes when the route changes.
- IPv6 dropped fragmentation at routers altogether: a datagram too large is discarded and the source is told.
- Those error messages are a protocol of this layer in their own right, and we read them when we come to diagnosing a network.

Summary

- An IP address is 32 bits and belongs to an interface. A mask splits it into a subnet part and an interface part.
- A subnet is what is left when the interfaces are detached: an island reachable without crossing a router, point-to-point links included.
- CIDR puts the boundary anywhere. A block is an interval, named by its lowest address and the prefix length.
- Blocks are allocated hierarchically and contiguously, so one prefix serves many customers — and a longer one carves out an exception.
- The header is 20 bytes: version, lengths, TTL, protocol, checksum, two addresses, and three fields that exist only for fragmentation.
- A datagram too large for the next link is cut by the router and rebuilt by the destination, at a price today's Internet prefers to avoid.